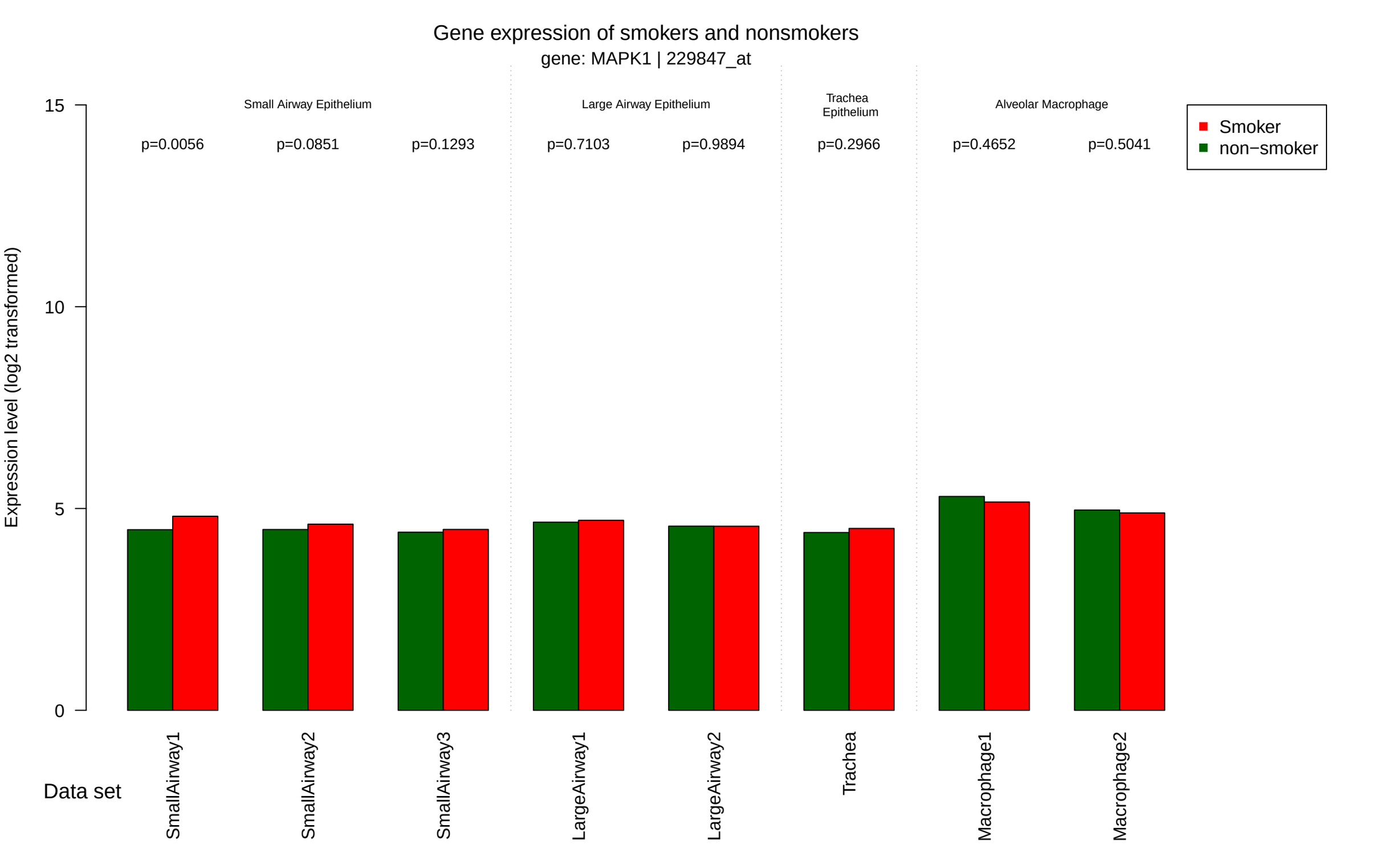
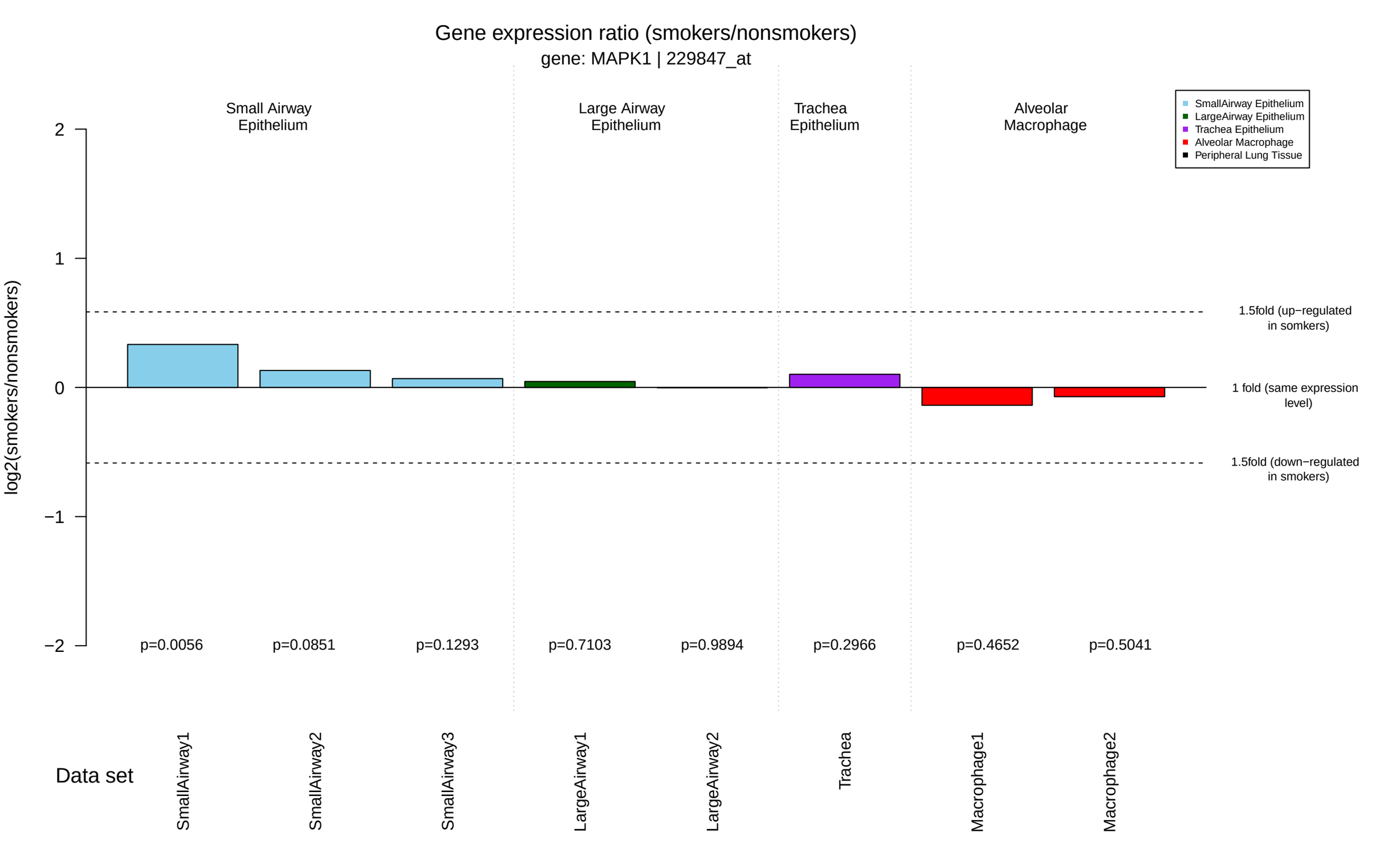
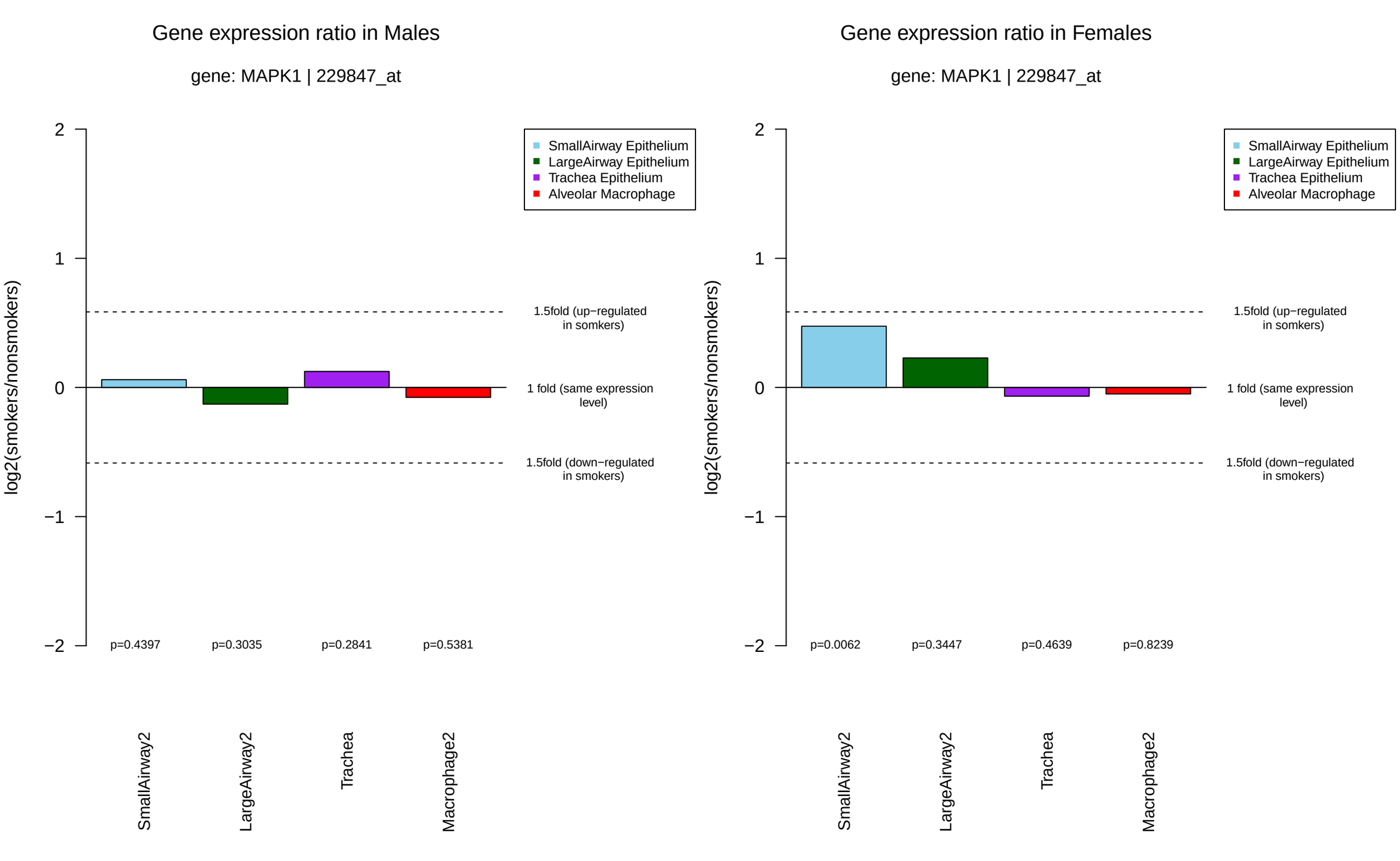
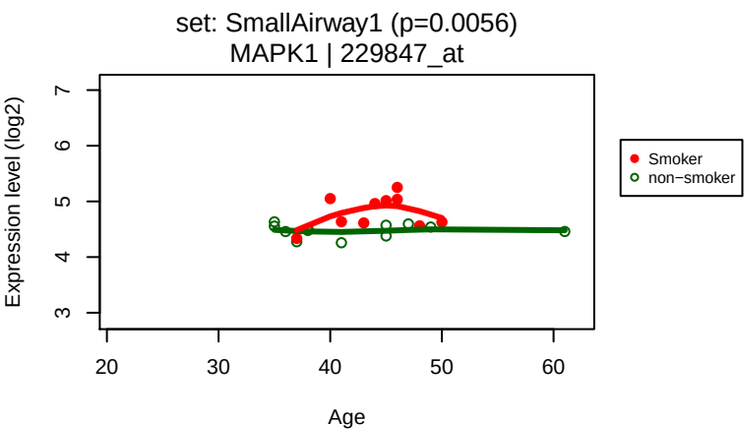
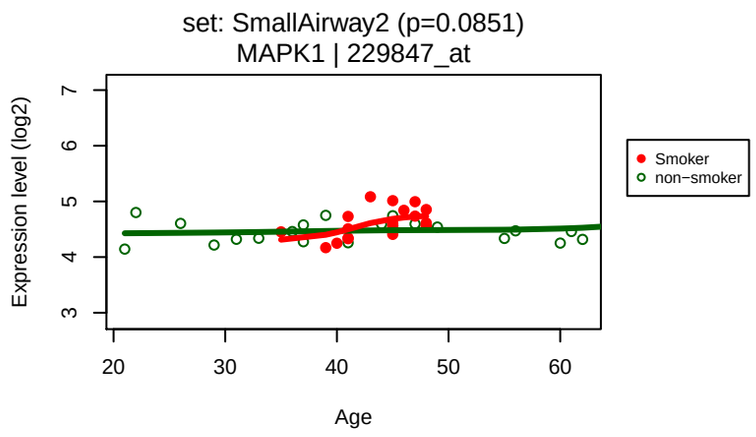
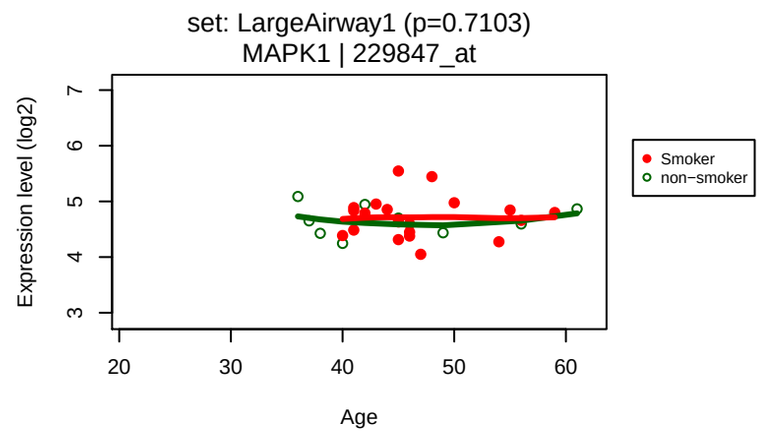

Gene expression of smokers and nonsmokers
gene: MAPK1 | 229847_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0056
p=0.0851
p=0.1293
p=0.7103
p=0.9894
p=0.2966
p=0.4652
p=0.5041
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

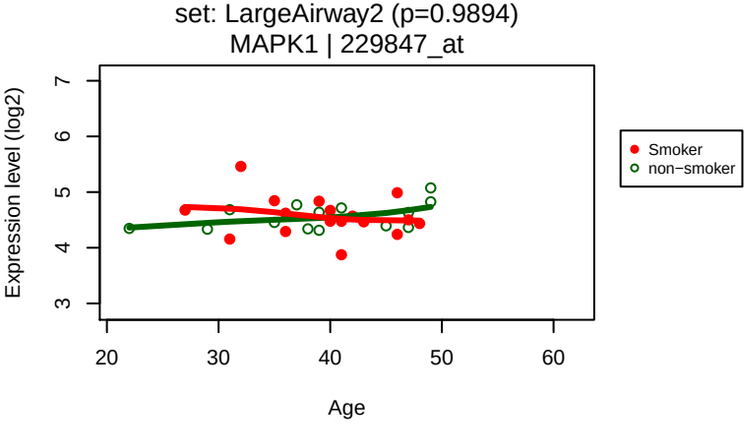

Gene expression ratio (smokers/nonsmokers)
gene: MAPK1 | 229847_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0056
p=0.0851
p=0.1293
p=0.7103
p=0.9894
p=0.2966
p=0.4652
p=0.5041
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

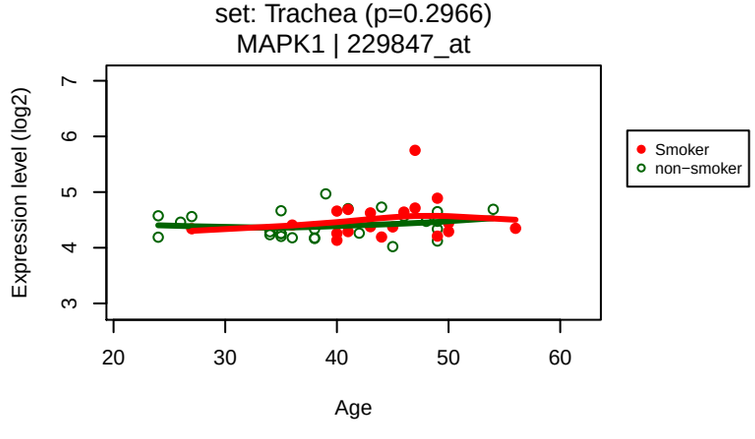

Gene expression ratio in Males
gene: MAPK1 | 229847_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4397
p=0.3035
p=0.2841
p=0.5381
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPK1 | 229847_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0062
p=0.3447
p=0.4639
p=0.8239
SmallAirway2
LargeAirway2
Trachea
Macrophage2

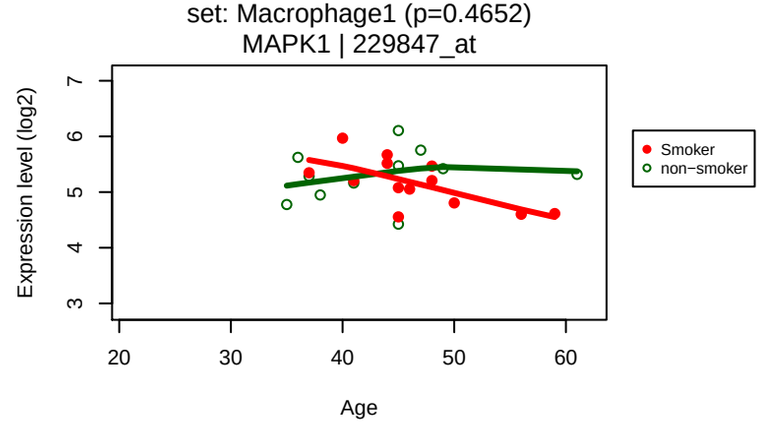

set: SmallAirway1 (p=0.0056)
MAPK1 | 229847_at
Expression level (log2)
Age
Smoker
non−smoker

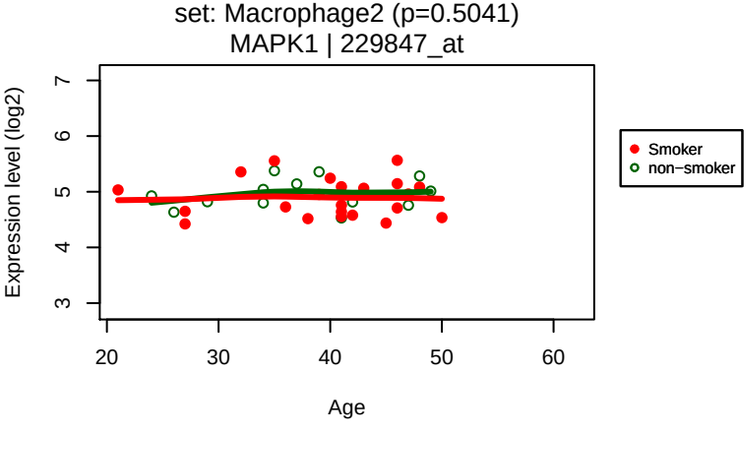

set: SmallAirway2 (p=0.0851)
MAPK1 | 229847_at
Expression level (log2)
Age
Smoker
non−smoker

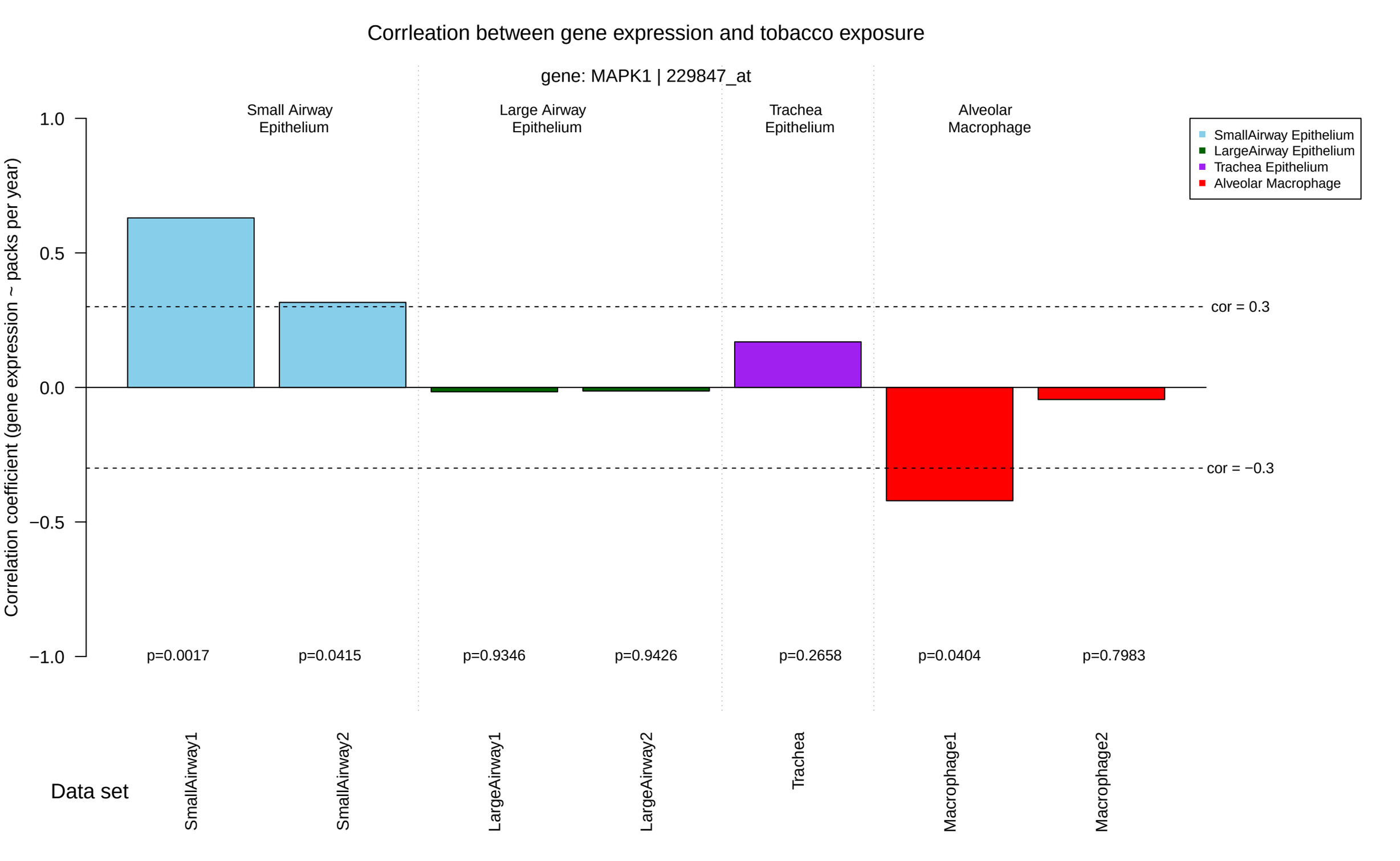

set: LargeAirway1 (p=0.7103)
MAPK1 | 229847_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9894)
MAPK1 | 229847_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2966)
MAPK1 | 229847_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4652)
MAPK1 | 229847_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5041)
MAPK1 | 229847_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPK1 | 229847_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0017
p=0.0415
p=0.9346
p=0.9426
p=0.2658
p=0.0404
p=0.7983
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2